

STAT 24620=FINM 34700, STAT 32950
Multivariate Data Analysis
Lecture 4: Factor Analysis

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Outline

- 1 Factor analysis model
- 2 Estimation and factor count selection
- 3 Factor rotation for interpretation
- 4 In-class notebook segment
- 5 Wrap-up

PCA → Factor analysis

- Lecture 1: multivariate normal setup and covariance structure.
- Lecture 2: PCA as orthogonal variance-maximizing projections.
- Lecture 3: PCA workflow + PPCA + practical interpretation.

Today: turn “directions” into a **latent variable model**.

- Separate common variation from variable-specific noise.
- Learn what is and is not identifiable in factor models.
- Estimate factors/loadings and rotate for interpretation.

Core factor analysis model

For observed vector $\mathbf{x}_i \in \mathbb{R}^p$ and latent factors $\mathbf{f}_i \in \mathbb{R}^m$ ($m < p$):

$$\mathbf{x}_i = \boldsymbol{\mu} + L\mathbf{f}_i + \boldsymbol{\epsilon}_i,$$

with assumptions

$$\mathbb{E}[\mathbf{f}_i] = \mathbf{0}, \quad \text{Cov}(\mathbf{f}_i) = I_m, \quad \mathbb{E}[\boldsymbol{\epsilon}_i] = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\epsilon}_i) = \Psi = \text{diag}(\psi_1, \dots, \psi_p),$$

and $\text{Cov}(\mathbf{f}_i, \boldsymbol{\epsilon}_i) = \mathbf{0}$.

- $L \in \mathbb{R}^{p \times m}$: loading matrix.
- Ψ : uniqueness (idiosyncratic variance) matrix.

Covariance decomposition

Under the model,

$$\Sigma = \text{Cov}(\mathbf{x}_i) = LL^\top + \Psi.$$

Interpretation by variable j :

$$\text{Var}(X_{ij}) = \underbrace{\sum_{k=1}^m \lambda_{jk}^2}_{\text{communality } h_j^2} + \underbrace{\psi_j}_{\text{uniqueness}}.$$

- **Communality** h_j^2 : variation explained by common factors.
- **Uniqueness** ψ_j : variable-specific noise/residual variance.

Conceptual goal of FA: explain covariance, not necessarily maximize total variance as PCA does.

Relation to PPCA from Lecture 3

PPCA model:

$$\mathbf{x}_i = W\mathbf{z}_i + \mu + \epsilon_i, \quad \mathbf{z}_i \sim \mathcal{N}(0, I_q), \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2 I_p),$$

FA model generalizes PPCA by replacing isotropic noise with diagonal noise:

$$\sigma^2 I_p \longrightarrow \Psi = \text{diag}(\psi_1, \dots, \psi_p).$$

- PPCA: equal uniqueness variance across variables.
- FA: variable-specific uniquenesses, often more realistic.

Takeaway: FA is more flexible for heterogeneous measurement noise.

Geometric vs probabilistic views of PCA and FA

PCA (geometric view)

- deterministic orthogonal directions;
- maximizes variance of projections;
- decomposes total variance;
- no explicit noise model.

Bridge: PPCA

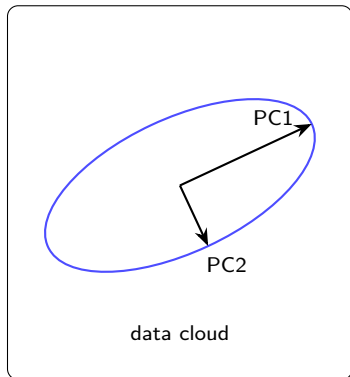
- special case of FA with $\Psi = \sigma^2 I_p$;
- maximum likelihood solution recovers PCA directions.

Factor Analysis (probabilistic view)

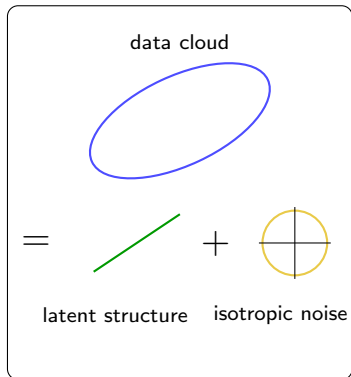
- stochastic latent factors;
- covariance: $\Sigma = LL^\top + \Psi$;
- separates common and idiosyncratic variation;
- includes variable-specific noise (Ψ).

PCA vs PPCA vs Factor Analysis

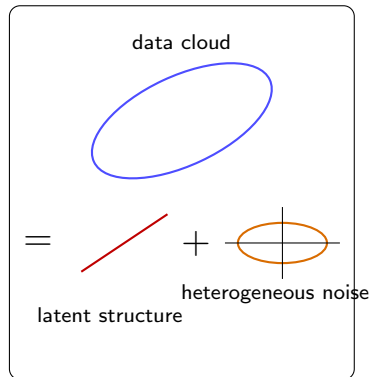
PCA



PPCA



FA



Another visual comparison between PCA and FA

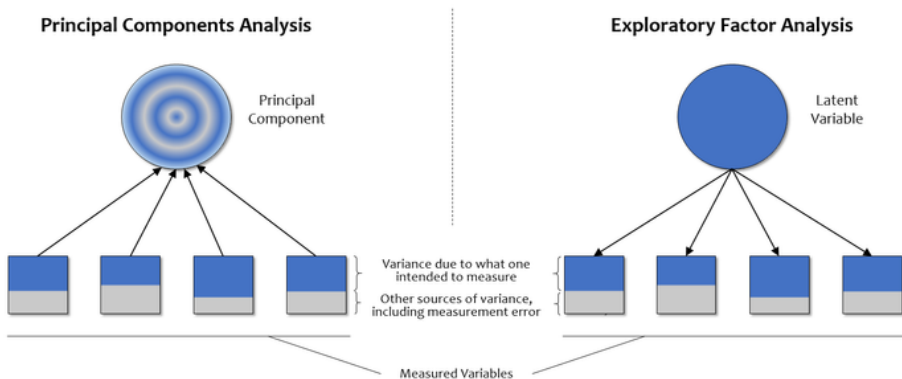


Figure: Figure from <https://community.jmp.com/t5/JMP-Blog/Principal-components-or-factor-analysis/ba-p/38347>

FA vs PCA: orthogonality and variance normalization

PCA

$$\mathbf{X}_c \approx \mathbf{Z}_m \mathbf{V}_m^\top$$

- Loading vectors are **orthonormal**:
 $\mathbf{V}_m^\top \mathbf{V}_m = \mathbf{I}$.
- Score coordinates are uncorrelated with different variances:

$$\text{Cov}(\mathbf{z}_i) = \text{diag}(\lambda_1, \dots, \lambda_m).$$

- Scale is absorbed into scores

Factor Analysis

$$\mathbf{x}_i = \boldsymbol{\mu} + \mathbf{L}\mathbf{f}_i + \boldsymbol{\epsilon}_i$$

- Columns of the loading matrix $\mathbf{L}$ are **not required to be orthogonal**.
- Latent factors are standardized and uncorrelated:

$$\text{Cov}(\mathbf{f}_i) = \mathbf{I}_m.$$

- Scale is absorbed into loadings.

Why diagonal Ψ is strong but useful

Assuming diagonal Ψ means uniqueness components are uncorrelated across variables.

- Good approximation when common dependence is captured by factors.
- Keeps parameter count manageable.

Parameter counting intuition:

- Unrestricted covariance: $p(p+1)/2$ parameters.
- FA model: $pm + p$ parameters before identifiability adjustments.

If $m \ll p$, FA can be a strong dimensionality reduction for covariance modeling.

Identifiability and rotational invariance

If $Q \in \mathbb{R}^{m \times m}$ is orthogonal ($Q^\top Q = I_m$), then

$$L\mathbf{f}_i = (LQ)(Q^\top \mathbf{f}_i),$$

and since $Q^\top \mathbf{f}_i$ still has covariance I_m , the covariance of $\mathbf{x}_i$ is unchanged:

$$LL^\top = (LQ)(LQ)^\top.$$

Consequence: factors/loadings are not unique without additional conventions.

- “Raw” factor directions are only identified up to rotation/sign/permutation.
- Interpretation needs a rotation criterion.

From covariance model to likelihood

From the factor model,

$$\text{Cov}(\mathbf{x}_i) = \Sigma = LL^\top + \Psi.$$

Goal: estimate (L, Ψ) so that Σ matches the sample covariance $\mathbf{S}$.

Under Gaussian assumptions:

$$\mathbf{f}_i \sim \mathcal{N}(0, I_m), \quad \epsilon_i \sim \mathcal{N}(0, \Psi) \Rightarrow \mathbf{x}_i \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma).$$

This allows likelihood-based estimation.

Maximum likelihood estimation for FA

Under multivariate normality, the log-likelihood is

$$\ell(L, \Psi) \propto -\log |\Sigma| - \text{tr}(\mathbf{S}\Sigma^{-1}), \quad \Sigma = LL^{\top} + \Psi.$$

Interpretation:

- choose Σ to approximate the sample covariance $\mathbf{S}$;
- decompose variability into
 - common structure: $LL^{\top}$,
 - variable-specific noise: Ψ .

Challenge:

- nonlinear parameterization $\Rightarrow$ no closed-form solution.

Estimation via EM: key idea

Latent factors $\mathbf{f}_i$ are unobserved.

Idea: treat latent factors as missing data.

E-step:

- estimate the latent factors given the observed data and current parameters.

M-step:

- update (L, Ψ) by fitting the model to these estimated factors.

Repeat until convergence.

Interpretation:

- alternates between estimating hidden structure and updating model parameters

This procedure is known as the EM algorithm.

Alternative estimation methods

Besides MLE, one can use:

- **The principal component / factor method:** approximate $S \approx LL^T + \Psi$ using PCA as solutions or initializations;
- **MINRES:** minimize residual off-diagonal covariance.

Remarks:

- do not require full distributional assumptions;
- often simpler but less statistically efficient than MLE.

How many factors m ? A practical FA workflow

Focus on two complementary diagnostics:

- **Scree plot / PA:** start from the eigenvalues of the sample *correlation* matrix.
 - Exactly the same steps as for PCA
- **Residual correlations after FA fit:** fit FA with a candidate m and check whether the off-diagonal entries of

$$R - \hat{R}$$

are small. Large residual blocks mean some common dependence is still unexplained.

Interpretation:

- too small $m \Rightarrow$ scree / parallel analysis suggests more factors and residual correlations remain structured;
- increase m until residual structure is mostly negligible.

Choose the smallest m that removes systematic residual correlation and still gives interpretable factors.

Rotation for interpretation

Because of rotational invariance, we can choose a rotation to make factors interpretable

Orthogonal rotation (eg., varimax):

$$L \rightarrow LQ, \quad \mathbf{f}_i \rightarrow Q^\top \mathbf{f}_i, \quad Q^\top Q = I_m$$

- factors remain uncorrelated: $\text{Cov}(\mathbf{f}_i) = I_p$;

Oblique rotation (eg., oblimin / promax):

$$L \rightarrow LQ, \quad \mathbf{f}_i \rightarrow Q^{-1} \mathbf{f}_i$$

- factors may become correlated: $\text{Cov}(\mathbf{f}_i) \neq I_p$;

Typical outcome:

- sparse loadings (each variable loads on few factors);
- easier interpretation (“market”, “sector”, etc.).

Illustrating factor rotation

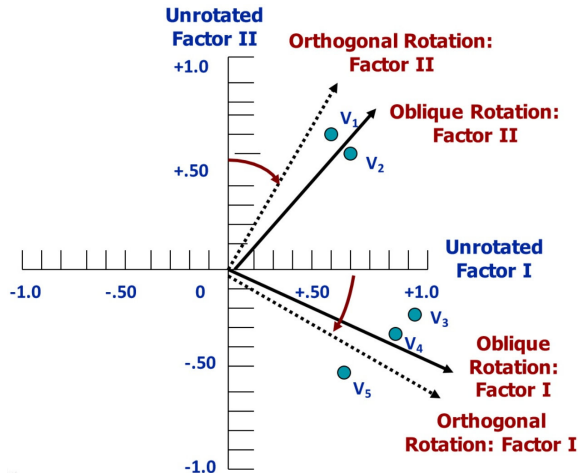


Figure: Source: <https://devopedia.org/factor-analysis>

Varimax criterion

For the loading matrix L and columns $l_{.k}$, Varimax seeks a rotation maximizing spread of squared loadings:

$$\max_Q \sum_{k=1}^m \left[\frac{1}{p} \sum_{j=1}^p \tilde{l}_{jk}^4 - \left(\frac{1}{p} \sum_{j=1}^p \tilde{l}_{jk}^2 \right)^2 \right], \quad \tilde{L} = LQ.$$

- Pushes loadings toward “large or near-zero” patterns.
- Helps create simple structure for communication.

A rotation example

Before

$$L = \begin{pmatrix} 0.60 & 0.67 \\ 0.53 & 0.67 \\ 0.42 & 0.71 \\ -0.57 & 0.71 \\ -0.46 & 0.74 \\ -0.35 & 0.78 \end{pmatrix}$$

After (varimax)

$$\tilde{L} = \begin{pmatrix} 0.90 & 0.10 \\ 0.84 & 0.14 \\ 0.79 & 0.24 \\ 0.00 & 0.90 \\ 0.16 & 0.86 \\ 0.26 & 0.81 \end{pmatrix}$$

After (promax)

$$\tilde{L} = \begin{pmatrix} 0.93 & 0.00 \\ 0.86 & 0.00 \\ 0.79 & 0.00 \\ -0.12 & 0.94 \\ 0.00 & 0.87 \\ 0.12 & 0.81 \end{pmatrix}$$

- Before: variables load on both factors.
- After: each variable loads mainly on one factor.

Rotation reveals a simple structure.

How to use factor analysis (in practice)

- 1 choose number of factors m ;
- 2 fit FA model;
- 3 rotate loadings for interpretation;
- 4 check and present residual correlations for model fit.

$$R_{\text{res}} = R - \hat{R}.$$

Inspect the largest off-diagonal residuals; persistent blocks suggest missing factors.

- **RMSR (root-mean-square-error) summary:**

$$\text{RMSR} = \sqrt{\frac{2}{p(p-1)} \sum_{i < j} r_{ij, \text{res}}^2}$$

computed from residual *correlations* gives one-number fit quality.

Goal: simple, interpretable structure with small residual correlations.

How do we get estimated factor scores?

After estimating Λ and Ψ , we often want a score estimate $\hat{\mathbf{f}}_i$ for each observation $\mathbf{x}_i$.

Regression scores:

$$\hat{\mathbf{f}}_i = \hat{\mathbb{E}}[\mathbf{f}_i \mid \mathbf{x}_i] = (I_m + \hat{L}^\top \hat{\Psi}^{-1} \hat{L})^{-1} \hat{L}^\top \hat{\Psi}^{-1} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}).$$

- This is the conditional mean of the latent factor under the Gaussian FA model.
- It gives the best linear predictor of $\mathbf{f}_i$ from the observed variables.
- In R, this is what `factanal(..., scores = "regression")` returns.

Interpretation:

- scores summarize where an observation lies along the latent factors;
- unlike PCA scores, these are model-based and depend on both L and Ψ ;

Notebook file: `Lecture4_demo.nb.html`

Common pitfalls in factor analysis

- Rotated factors are not uniquely defined (rotation invariance).
- Rotation does not necessarily improve interpretation.
- More factors improve fit, but may reduce interpretability.
- Factors capture statistical patterns, not necessarily causal mechanisms.

Interpret factors cautiously.

Lecture 4 summary

- ① Factor analysis models covariance through a small number of latent factors plus idiosyncratic noise.
- ② Communalities capture shared variation; uniquenesses capture variable-specific noise.
- ③ PCA is a variance decomposition; factor analysis is a generative model for covariance.
- ④ Estimation balances fit and structure.
- ⑤ Rotation improves interpretability without changing model fit.
- ⑥ In practice, factor models provide an interpretable structure for multivariate data.

Suggested reading

- Johnson & Wichern (6th Edition), Chapter 9.
- Detailed derivations on EM for factor analysis:
<https://www.math.hkbu.edu.hk/~hpeng/Math3806/EM-factor.html>